PHYSICS-INFORMED XGBOOST FOR HIGH-RESOLUTION METEOROLOGICAL DOWNSCALING: ANCHORING MACHINE LEARNING WITH TOPOGRAPHICAL DYNAMICS

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ABSTRACT

We present a physics-informed machine learning framework for downscaling coarse meteorological baselines (e.g., 9km European Centre for Medium-Range Weather Forecasts Integrated Forecasting System (ECMWF IFS)) to hyper-local 30m resolution. While pure data-driven approaches often struggle to conserve physical laws across complex topography, our architecture anchors PI-XGBoost predictions using explicit physical constraints derived from Digital Elevation Models (DEM). Specifically, we introduce deterministic physical anchors for temperature (environmental lapse rate) and pressure (barometric scale height). Initial results explore how formulating this as physics-regularized residual learning guides the model toward physically plausible predictions in complex terrain, establishing a foundation for sparse-station supervised residual downscaling.

1 Introduction

Spatial downscaling of numerical weather prediction models remains a persistent challenge in atmospheric sciences. Global reanalysis datasets, such as ERA5-Land, provide continuous, reliable estimates of meteorological variables (t, p, u_{10}, v_{10}) but at a coarse spatial resolution ($\sim$9km) that smooths over critical topographic variations.

While deep learning approaches have gained traction for downscaling tasks, they are notoriously data-hungry and computationally expensive. Furthermore, supervised learning approaches face immense difficulties in scenarios where dense, high-resolution target grids are unavailable. To address this, we propose a sparse-station supervised residual downscaling methodology utilizing Physics-Informed XGBoost (PI-XGBoost). This approach embeds explicit atmospheric physics into the modeling pipeline, allowing the algorithm to dynamically correct coarse ERA5-Land baselines using 30m Copernicus Digital Elevation Models (DEM), utilizing sparse observational stations rather than relying on dense historical ground-truth grids, which are themselves often limited in accuracy, resolution, or availability. This project showcases a novel way to combine a DEM, a station network, and a physics-based loss function into a downscaling task to provide an easy-to-advance proof of concept.

2 Model Formulation

Our PI-XGBoost framework is designed to predict one dense target variable at a time (t, p, u_{10}, or v_{10}). Rather than predicting the absolute value from scratch, the model operates on a physics-regularized residual learning paradigm anchored by standard atmospheric formulas.

2.1 Inputs and Data Sources

The model utilizes three primary data streams:

- **Coarse Baseline:** ERA5-Land reanalysis data providing macro-level t, p, u_{10}, v_{10}.
- **Topography:** Copernicus DEM at 30m resolution, used to compute the elevation delta ($\Delta z = z_{30m} - z_{coarse}$).
- **Sparse Reference Targets:** FMI synoptic and weather stations (t, p, u_{10}, v_{10}) used to supervise the residual downscaling and evaluate performance.

Using ERA5-Land reanalysis for training acts as a highly stable proxy, enabling direct, seamless inference of this model using live operational data from the ECMWF IFS, the current state-of-the-art numerical model. Although we actively utilize IFS data to validate real-time inference within our wider system architecture, detailed evaluation of the operational real-time pipeline remains beyond the scope of this initial study.

2.2 Physics-Informed Anchors

To constrain the PI-XGBoost algorithm, we construct deterministic physical anchors based on the elevation delta. The model's objective is strictly limited to predicting the non-linear, sub-grid residual $\mathcal{R}$ that physical formulas cannot capture (e.g., land-use anomalies, local shading).

The final 30m prediction $\hat{Y}$ is defined as:

$$\hat{Y}_{30m} = Y_{anchor}(X_{coarse}, \Delta z) + \text{XGBoost}_{residual}(X_{coarse}, \text{DEM}) \tag{1}$$

Pressure Anchor: Pressure is anchored using the standard barometric formula, constrained by the atmospheric scale height H [3, 5]:

$$P_{anchor} = P_{coarse} \exp\left(-\frac{\Delta z}{H}\right) \tag{2}$$

Temperature Anchor: Temperature adjustments are anchored linearly via the environmental lapse rate Γ, derived from the International Standard Atmosphere (≈ -0.0065°C/m) [4, 6]:

$$T_{anchor} = T_{coarse} + \Gamma \Delta z \tag{3}$$

3 Initial Results

By formulating the task as physics-regularized residual learning, the PI-XGBoost algorithm is specifically tasked with predicting localized deviations from the physical anchors. Because the model is computationally lightweight, training and inference over regional bounding boxes can be executed efficiently on standard laptop hardware. Validation against sparse FMI station data indicates that the PI-XGBoost output provides a physically constrained high-resolution field that offers a structural improvement over the coarse baseline. This enhancement is illustrated by the time-series comparisons for wind speed and air temperature shown in Figures 1 and 2, respectively.

Ongoing evaluations focus on quantifying how these explicit physical constraints aid overall convergence stability and out-of-sample generalization. The learning curves for temperature, pressure, and the u_{10} wind component, presented in Figure 3, demonstrate the practical viability of this approach. Future refinements to the model architecture, hyperparameter tuning, and data preprocessing are expected to further improve predictive performance.

4 Discussion and Future Advancements

The PI-XGBoost architecture demonstrates that incorporating physical equations into the residual learning pipeline enables effective sparse-station supervised residual downscaling. It acts as a guardrail, aiming to prevent the model from wildly violating fundamental atmospheric constraints when projecting into unobserved, high-resolution topographical spaces.

Having established this concept with a highly efficient, laptop-runnable framework, our subsequent work will transition toward Physics-Informed Neural Networks (PINNs) [7]. While PI-XGBoost excels at tabular mapping, PINNs will enable the handling of coupled partial differential equations. This transition will be particularly crucial for downscaling wind vectors (u_{10}, v_{10}), where explicit mass conservation and Navier-Stokes constraints must be enforced

Figure 1: Time-series comparison of the v_{10} wind speed component. The PI-XGBoost prediction is evaluated against the ERA5-Land coarse baseline and the ground-truth FMI station measurements.

across the 3D spatial grid—a task better suited for differentiable deep learning architectures. Additionally, a problem-specific neural network architecture is currently under development.

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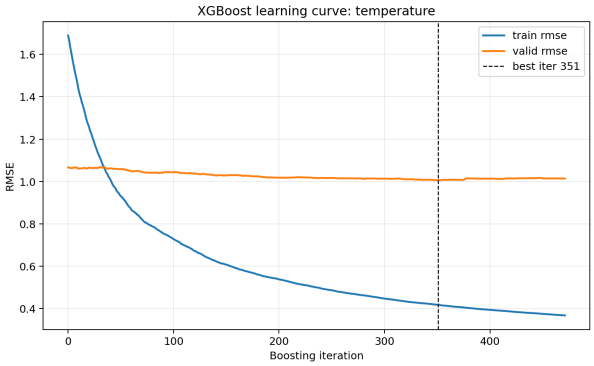
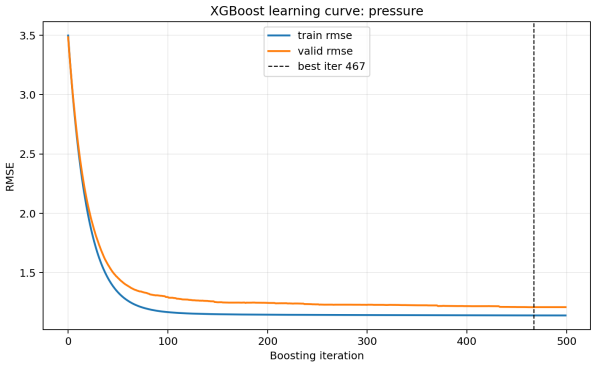
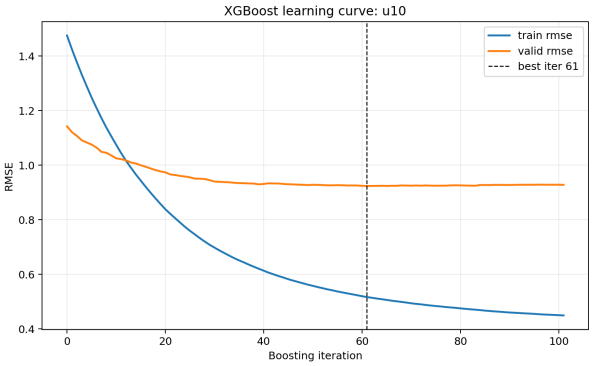
Figure 2: Time-series comparison of air temperature. The PI-XGBoost prediction demonstrates its structural improvements over the ERA5-Land coarse baseline when validated against ground-truth FMI observations.



(a) Temperature



(b) Pressure



(c) u_{10} Wind Speed

Figure 3: Learning curves for the PI-XGBoost model predicting (a) air temperature, (b) atmospheric pressure, and (c) the u_{10} wind speed component.

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